

EduFlow

Online Learning Management System

Use-Case Diagrams

Based on System Requirements Specification — Version 2.1

Document	EduFlow Use-Case Diagrams
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1. Introduction

This document presents the Use-Case Diagrams for the EduFlow Online Learning Management System (LMS), developed as part of the Software Engineering 2 course project. These diagrams are derived directly from the Software Requirements Specification (SRS) Version 2.1 and provide a visual representation of the functional interactions between the system's actors and its core use cases.

EduFlow is a microservices-based web platform that enables instructors to create and publish courses, and students to discover, enroll in, and learn from them. The system is secured via JWT-based authentication, built on Spring Boot, and served through a React frontend.

System Actors

Actor	Role	Description
Admin	Platform Administrator	Full system control: user management, course moderation, analytics, and broadcasting.
Instructor	Course Creator / Educator	Creates and publishes courses after Admin approval. Views enrollments and earnings.
Student	Learner	Browses, enrolls in, and pays for courses. Tracks progress and submits reviews.
Public	Unauthenticated Visitor	Can browse and view published course details only. No enrollment or payment access.
Stripe (External)	Payment Gateway	Processes card transactions and sends webhook events to confirm payment status.
System (Internal)	Microservice Bus	Represents internal Feign calls that trigger notifications between services.

Diagrams Overview

#	Diagram Title	Actors	Domain
1	Authentication & User Management	Public, Student, Instructor, Admin	Auth, JWT, Profiles
2	Course Management	Public, Student, Instructor, Admin	CRUD, Enrollment, Progress
3	Payments & Reviews	Student, Instructor, Admin, Stripe	Stripe, Ratings, Replies
4	Notifications & Admin Oversight	Student, Instructor, Admin, System	Alerts, Moderation

2. Authentication & User Management

Diagram 1 of 4 | Domains: auth-service, user-service

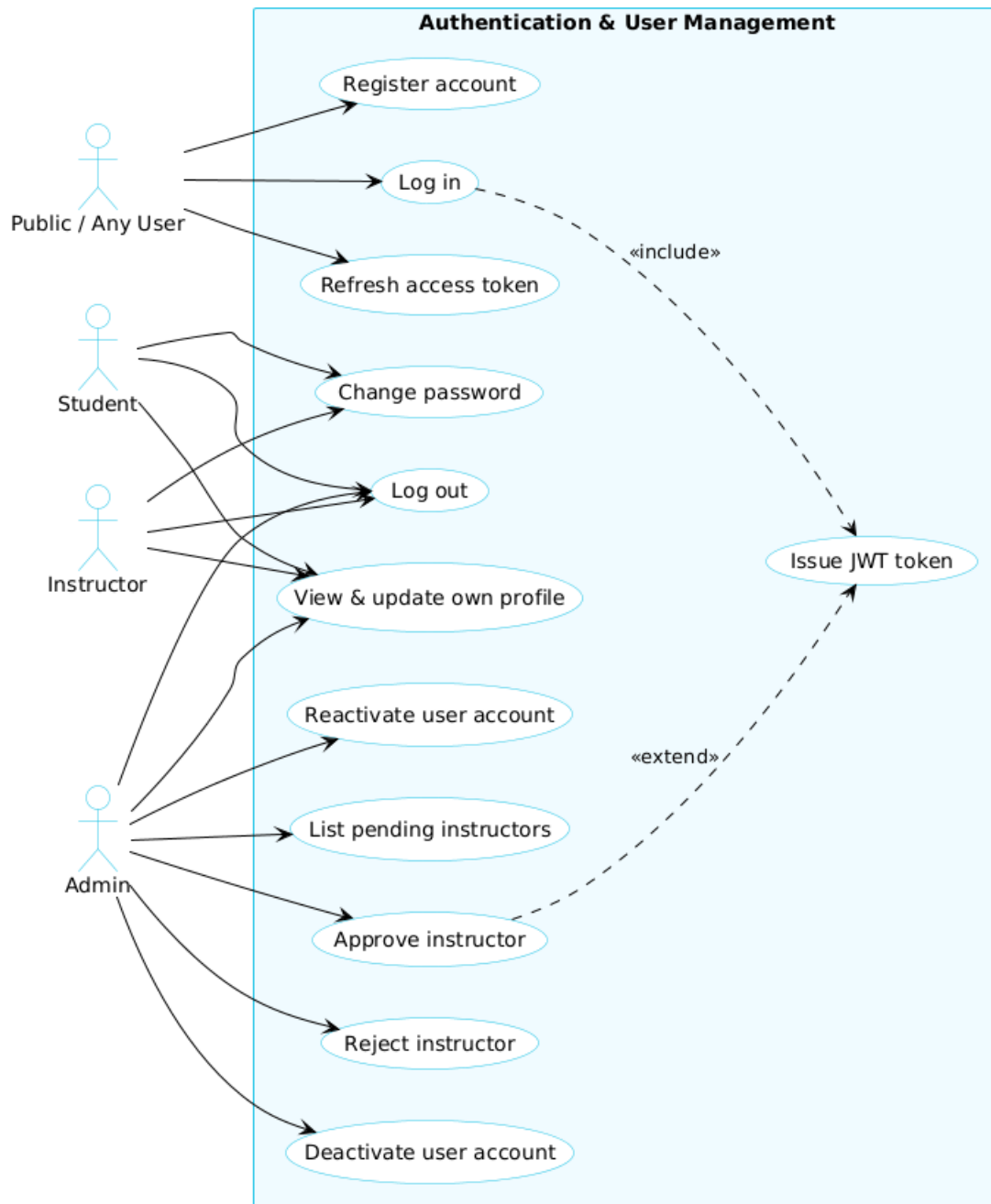
Overview: This diagram covers all actors interacting with the authentication and user-management domain. It models the registration and login flows, JWT token issuance and refresh, secure logout, profile management, and the instructor approval workflow. The <<include>> relationship from 'Log in' to 'Issue JWT token' reflects the mandatory token issuance on every successful login. The <<extend>> on 'Approve instructor' reflects that approval optionally triggers token-related state changes.

Actors Involved Public / Any User Student Instructor Admin	Use Cases Count Total use cases in this diagram: 12 Key Relationships Log in <<include>> Issue JWT token Approve instructor <<extend>> Issue JWT token
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Diagram



Figure 1 — Authentication & User Management Use-Case Diagram



3. Course Management

Diagram 2 of 4 | Domain: course-service

Overview: This diagram covers the full lifecycle of a course on EduFlow — from creation (saved as DRAFT) through lesson management, publishing, and public discovery. Students can browse, search, enroll in free courses directly, mark lessons complete, and track their overall progress. The Instructor manages their own course content and can view enrolled students. The Admin has override capabilities to unpublish, delete, or feature any course.

Actors Involved Public Student Instructor Admin	Use Cases Count Total use cases in this diagram: 12 Key Relationships Create course (draft) <<extend>> Manage lessons Publish course <<include>> Browse published courses
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Diagram



Figure 2 — Course Management Use-Case Diagram



4. Payments & Reviews

Diagram 3 of 4 | Domains: payment-service, review-service

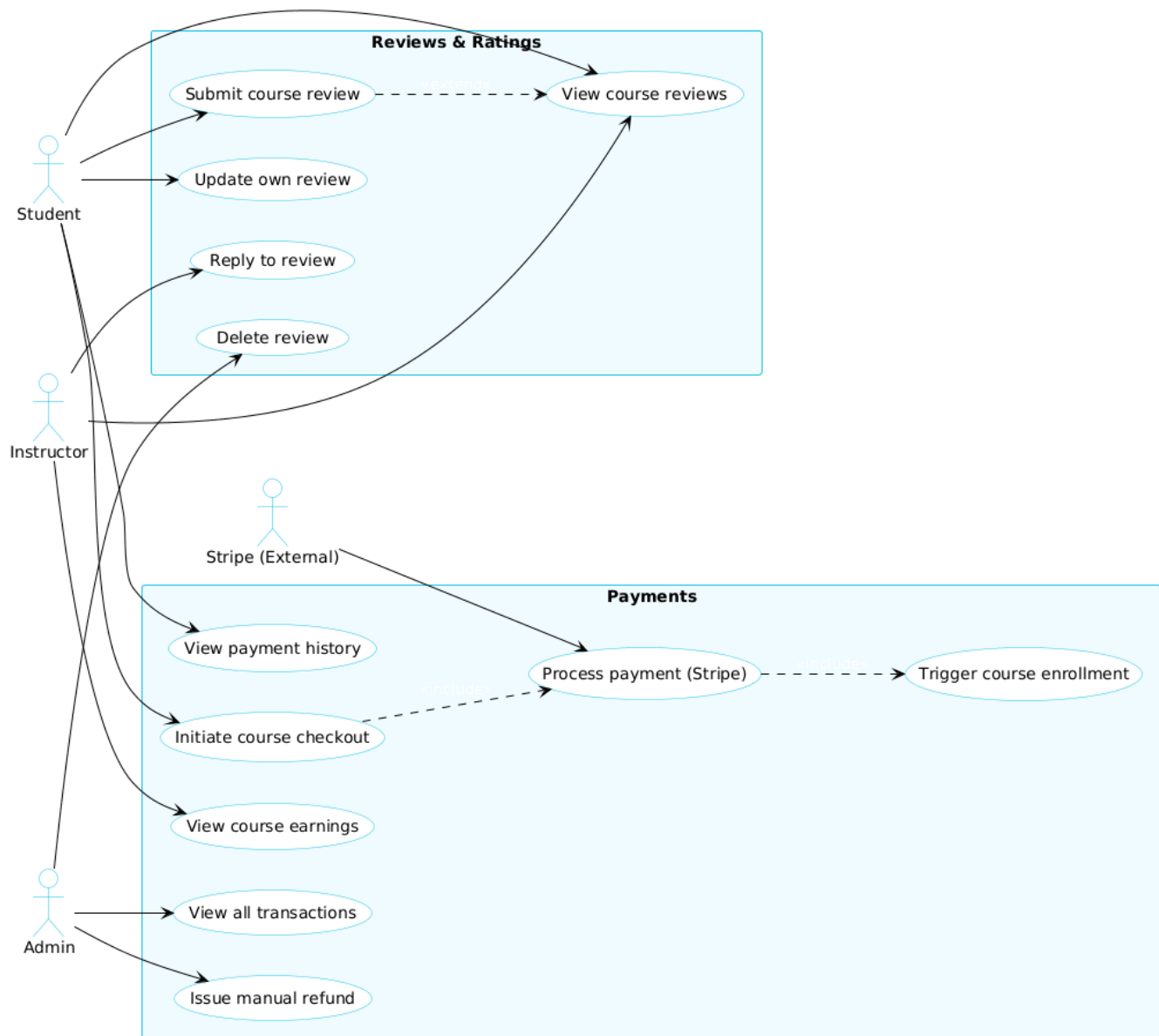
Overview: This diagram models the payment journey (Stripe checkout, webhook processing, enrollment trigger) and the complete reviews and ratings lifecycle. Stripe is represented as an external actor that processes card transactions and confirms payment status via webhooks. Enrollment is automatically triggered by a successful payment through an internal Feign call. Only enrolled students may submit or update reviews. Instructors can publicly reply to reviews on their own courses, and Admins can delete any review.

Actors Involved Student Instructor Admin Stripe (External)	Use Cases Count Total use cases in this diagram: 12 Key Relationships Initiate checkout <<include>> Process payment (Stripe) Process payment <<include>> Trigger course enrollment Submit review <<extend>> View course reviews
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Diagram



Figure 3 — Payments & Reviews Use-Case Diagram



5. Notifications & Admin Platform Oversight

Diagram 4 of 4 | Domains: notification-service, admin oversight

Overview: This diagram covers in-app notification delivery and administrative platform oversight. Notifications are triggered by internal Feign calls from other services (represented by the 'System (Internal)' actor) covering enrollment confirmations, instructor approval/rejection alerts, and new review alerts. All authenticated users can view and mark their notifications as read. The Admin can additionally send broadcast notifications to all users or to a specific role. The Admin Oversight section consolidates all cross-domain administrative capabilities in one place.

Actors Involved	Use Cases Count
Student Instructor Admin System (Internal)	Total use cases in this diagram: 13
	Key Relationships Receive enrollment confirmation <<extend>> View in-app notifications Receive approval/rejection alert <<extend>> View in-app notifications Receive new review alert <<extend>> View in-app notifications

Diagram



Figure 4 — Notifications & Admin Platform Oversight Use-Case Diagram

